

Chapter 5 — Pipelines, automation, and visualization

Single-notebook scripts do not scale across teams

Learning objectives

- Explain why pipelines prevent leakage
- Contrast dplyr/tidyr with pandas

R: dplyr and tidyr

- In R workflows
- Example 5.58 shows filter and select patterns analogous to pandas
- Teams standardized on Python may still encounter R extracts

sklearn pipelines

- Pipelines chain imputers, scalers, encoders
- This prevents statistics computed on validation data from leaking into features
- Serialize pipelines with the model for consistent deployment preprocessing

Automation: Featuretools and AutoML

- Featuretools performs deep feature synthesis on relational tables
- AutoML platforms wrap search over imputers, encoders, and model families
- Automation accelerates baselines but still requires domain review to drop leaky or

Example 5.74 — Deep feature synthesis on transactions

- Example 5.74 — hands-on module
- Example 5.74 shows automated aggregates across related tables
- Explore the chapter example module
- View files: ``modules/chapter5/example74/``

Visualization for cleaning

- Box plots expose outlier tails
- Visual checks complement rule-based flags from earlier parts and align with exploratory

Takeaways

- Pipelines encode reproducible preprocessing
- Automation proposes features; humans curate them
- Visual diagnostics catch defects rules miss

Next

- Complete the quiz for this part
- Continue to challenges, best practices, and governance habits